

CONCEPT OF MODELING BICM2210

MODULE DETAILS

Course Location : Lesotho
 Department : Faculty of Information & Communication Technology
 Program Name : BSc in Information Technology
 BSc in Software Engineering with Multimedia
 Semester : 4
 Credits : 10
 Status : Major
 Contact hours : 4 hours (2 hours lecture + 2 hours tutorial)
 No. of weeks : 14 teaching weeks + 1 Final examination week + 1 week Midterm Break
 Teaching Pattern : Lectures and Tutorial
 Pre-requisite : NIL
 Principal Lecturer : Mr Bhila
 Lecturer(s) : Mr Moorosi

Prepared by : T. Moorosi Checked by : T. BHILA
 Signature : [Signature] Date 29/01/26 Signature : [Signature] Date 29/01/26

This document comprises the following:

- Essential Information
- Specific Module Information
- Module Rules & Regulations
- Grades
- Plagiarism
- Module Introduction
- Module Aims & Objectives
- Learning Outcome
- Specific Generic Learning Skills
- Syllabus + Lecture Outline
- References
- Assessment Schedule
- Assessment Criteria
- Specific Criteria

Other documents as follows will be issued to you on an ongoing basis throughout the semester:

- Handouts for Assignments
- Submission Requirements + Guidelines

1.0 ESSENTIAL INFORMATION

- All modules other than electives are 'significant modules'
- As an indicator of workload one credit carries an additional 2 hours of self study per week. For example, a module worth 10 credits require that the student spends an additional 20 hours per

week, either reading, completing the assignment or doing self directed research for that module.

- Submission of ALL assignment work is compulsory in this module. A student cannot pass this module without having to submit ALL assignment work by the due date or an approved extension of that date.
- All assignments are to be handed on time on the due date. Students will be penalized 10 percent for the first day and 5 percent per day thereafter for late submission (a weekend or a public holiday counts as one day). Late submission, after the date Board of Studies meeting will not be accepted.
- Due dates, compulsory assignment requirements and submission requirements may only be altered with the consent of the majority of students enrolled in this module at the beginning/early in the program.
- Extensions of time for submission of assignment work may be granted if the application for extension is accompanied by a medical certificate.
- Overseas travel is not an acceptable reason for seeking a change in the examination schedule.
- Only the Head of School can grant approval for extension of submission beyond the assignment deadline.
- Re-submission of work can only receive a 50% maximum pass rate.
- Supplementary exams can only be granted if the level of work is satisfactory **AND** the semester work has been completed.
- Harvard referencing and plagiarism policy will apply on all written assignments.

2.0 SPECIFIC MODULE INFORMATION

- Attendance rate of 80% is mandatory for passing module.
- All grades are subject to attendance and participation.
- Absenteeism at any scheduled presentations will result in zero mark for that presentation.
- Visual presentation work in drawn and model form must be the original work of the student.
- The attached semester program is subject to change at short notice.

3.0 MODULE RULES AND REGULATIONS:

Assessment procedure:

- These rules and regulations are to be read in conjunction with the UNIT AIMS AND OBJECTIVES
- All assignments/projects must be completed and presented for marking by the due date.
- Marks will be deducted for late work and invalid reasons.
- All assignments must be delivered by the student in person to the lecturer concerned. No other lecturer is allowed to accept students' assignments.
- All tests/examinations are compulsory.
- Students must sit for the test/examination on the notified date.
- Students are expected to familiarise themselves with the test/examination timetable.
- Students who miss a test/examination will not be allowed to pass.
- Any scheduling of tutorials, both during or after lecture hours, is TOTALLY the responsibility of each student. Appointments are to be proposed, arranged, confirmed, and kept, by each student. Failure to do so in a professional manner may result in penalty of grades. Tutorials WITHOUT appointments will also NOT be entertained.
- Note that every assignment is given an ample time frame for completion. This, together with advanced information pertaining deadlines gives you NO EXCUSE not to submit assignments on time.

4.0 GRADES

All modules and assessable projects will be graded according to the following system. With respect to those units that are designated 'Approved for Pass/Fail' the grade will be either PX or F:

<i>Grade</i>	<i>Numeric Grade</i>	<i>Description</i>
90 – 100	A+	Pass with Distinction
85 – 89	A	
80 – 84	A-	
75 – 79	B+	Pass with Credit
74 – 70	B	
65 – 69	B-	
60 – 64	C+	Pass
55 – 59	C	
50 – 54	C-, PX	
0 – 49	F	Fail

EXP	Exempted
PP	Pass Provisional with extra work needed
PX	Pass after extra work is given and passed
X	Ineligible for assessment due to unsatisfactory attendance
D	Deferred
W	Withdraw

5.0 PLAGIARISM, COPYRIGHT, PATENTS, OWNERSHIP OF WORK: STUDENT MAJOR PROJECT, THESES & WORKS

See LIMKOKWING, HIGH FLYERS HANDOUT, pg 10.

6.0 MODULE INTRODUCTION

The module aims to provide students with an understanding of the concepts and principles of modeling. The module will also serve as a platform where advance and comprehensive 3D modeling technologies are introduced.

7.0 MODULE AIMS AND OBJECTIVES

To provide fundamental skills in digital media techniques, where a 2 dimensional design is explored in a 3 dimensional environment.

8.0 LEARNING OUTCOME

Upon completion of this module, the student will be able to:

- Explain 3 dimensional modeling
- Identify computer aided design application area
- Differentiate techniques used in constructing 3D models
- Examine use of NURBS in 3D modeling
- Explain the application area of CSG

9.0 SPECIFIC GENERIC LEARNING SKILLS

Upon completion of this module, student will acquire skills in:

- Understanding
- Comprehension
- Analysis
- Application
- Synthesis

10.0 UNIT SYLLABUS + LECTURE OUTLINE:

Week: 1

LECTURE 1: INTRODUCTION TO 3 DIMENSIONAL

Lecture Synopsis:

- Coordinate System
- Vertex, Faces and Objects
- Concept of wire frame, surface & solid modelling
- Orthographic Projection
- Axonometric Projection
- Oblique Projections
- Perspective Projections

Handout:

Tutorials

Week: 2 & 3

LECTURE 2: COMPUTER AIDED DESIGN

Lecture Synopsis:

- History
- Application
- Capabilities
- Technologies
- Operations

Handout:

Tutorials

Week: 4 & 5

LECTURE 3: CONSTRUCTIVE SOLID GEOMETRY

Lecture Synopsis:

- CSG
- Operation
- Application

Handout:

Tutorials

Week: 6 & 7

LECTURE 4: LOFTING, BOOLEAN OPERATIONS

Lecture Synopsis:

- Loft Objects
- Creating Shapes for Lofting
- Boolean Operations

Handout:

Tutorials

Week: 8 & 9

LECTURE 5: LIGHTING

Lecture Synopsis:

- 4.1 Property of Light
- 4.2 Natural and Artificial Light
- 4.3 Creating Light Objects
- 4.4 Omni Light
- 4.5 Spotlights & Free Spotlight
- 4.6 Setting Light colour
- 4.7 Setting Shadow

Handout:

Tutorials

Week: 10 & 11

LECTURE 5: NON-UNIFORM RATIONAL B-SPLINE

- NURB definition
- History
- Practical usage

Handout:

Tutorials

Week: 12

MID TERM BREAK WEEK

Week: 13 & 14

LECTURE 6: SOLID MODELING

Lecture Synopsis:

- Definition
- History
- Basic theoretical concept
- Practical usage

*Handout:
Tutorials*

Week: 15

LECTURE 7: PARTICLE SYSTEMS

Lecture Synopsis:

- Particle systems for realistic photo-rendering
- Creating simple particle systems
- Definition of Space Warp, concept & applications
- Space warp that affects particles or geometry

*Handout:
Tutorials*

Week: 16 & 17

FINAL EXAMINATION WEEK

11.0 REFERENCES

1. Breiud, F., 2024 Creo Parametric 2025: 1st ed. Scribd; Packet Publishing.
 2. Juohen, M., 2024. 3DS Max 2025 guide for beginners: 2nd ed. Amazon: paperback
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12.0 ASSESSMENT SCHEDULE

ASSIGNMENT DESCRIPTION	ISSUE DATE	DUE DATE	%
ASSIGNMENT 1:	WEEK 5	WEEK 8	20%
ASSIGNMENT 2:	WEEK 8	WEEK 11	20%
ASSIGNMENT 3:	WEEK 11	WEEK 14	20%
FINAL EXAMINATION:	WEEK 16	WEEK 16	40%
TOTAL			100%

Handwritten signature and date:
29/01/2026

13.0 ASSESSMENT CRITERIA

Process of grading and criteria used to determine the grades, passes and high distinctions.

14.0 SPECIFIC CRITERIA

- Each assignment will be handed out with the project brief and will vary, depending on the teaching and learning objectives of the specific assignment.
- Each student will receive a completed assessment sheet back with their marks, thereby giving student feedback on each set criterion and the project as a whole.
- Marks for each project will be posted on the Bulletin Board with student number within 2 weeks of hand-in date.